Math 54 — Week 3

Solutions

P1.

Idea. The columns are independent exactly when there is a pivot in every column.

1. Row reduce:
$$\begin{bmatrix} 1 & -3 & 2 \\ 3 & -8 & 5 \\ -2 & 7 & -3 \end{bmatrix} \xrightarrow[R_3+2R_1]{R_2-3R_1} \begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$
2. This echelon form has a pivot in each of the three columns, so $A\mathbf{x} = \mathbf{0}$ has no free variable and the columns are independent.

P2.

Idea. Row reduce with h as a letter; the set is dependent exactly when some column has no pivot.

1. Put the vectors into the columns and row reduce:
$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ -1 & 1 & h \end{bmatrix} \xrightarrow[R_3+R_1]{R_2-2R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 3 & h+1 \end{bmatrix} \xrightarrow{R_3-3R_2} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & h-5 \end{bmatrix}$$
2. If $h \neq 5$, every column has a pivot: independent. If $h = 5$, column 3 has no pivot: dependent.
3. For $h = 5$, take $x_3 = 1$. Row 2 gives $x_2 = -2x_3 = -2$, and row 1 gives $x_1 = -2x_2 - x_3 = 3$. So
$$3\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} - 2\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 3-4+1 \\ 6-10+4 \\ -3-2+5 \end{bmatrix} = \mathbf{0}.$$

P3.

Idea. Use the inspection rules of 1.7: too many vectors (Thm. 8), the zero vector (Thm. 9), or two vectors that are multiples.

1. (a) Dependent: three vectors in $\mathbb{R}^2$ (Thm. 8).
2. (b) Dependent: $(2, -4, 6) = -2(-1, 2, -3)$.
3. (c) Dependent: the set contains $\mathbf{0}$ (Thm. 9).
4. (d) Independent. Neither vector is $\mathbf{0}$, so if one were a multiple of the other, each would be a nonzero multiple of the other; then it is enough to check one direction. Every multiple $c(1, 2, 0)$ has third entry 0, while $(0, 1, 3)$ has third entry 3. So neither is a multiple of the other.

P4.

Idea. The composition is linear, so its columns are the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ followed through both moves.

1. The shear by 2 sends (x_1, x_2) to $(x_1 + 2x_2,\ x_2)$, and the reflection through the x_2-axis sends (x_1, x_2) to $(-x_1,\ x_2)$.
2. $\mathbf{e}_1 = (1, 0)$: the shear gives $(1, 0)$, the reflection gives $(-1, 0)$. $\mathbf{e}_2 = (0, 1)$: the shear gives $(2, 1)$, the reflection gives $(-2, 1)$. So the standard matrix is
$$\begin{bmatrix} -1 & -2 \\ 0 & 1 \end{bmatrix}.$$
3. It is already in echelon form with pivots -1 and 1 in both columns and both rows, so T is one-to-one and onto (Thm. 12).

P5.

Idea. Solve $A\mathbf{x} = \mathbf{b}$ by row reduction; then read one-to-one and onto from the pivots of A.

1. Row reduce:
$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 2 & -3 & 5 & 0 \end{array}\right] \xrightarrow{R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -1 & 2 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 + 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$
2. x_3 is free, $x_1 = 3 - x_3$ and $x_2 = 2 + x_3$, so
$$\mathbf{x} = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$
3. Column 3 of A has no pivot, so T is not one-to-one; row 3 has no pivot, so T is not onto (Thm. 12).

P6.

Idea. Test $T(\mathbf{0})$ first; if a map is given by a formula with no constant terms, write it as $A\mathbf{x}$.

1. $T(0, 0) = (0, -1) \neq \mathbf{0}$, so T is not linear.
2. Entry by entry, S has coefficients $(0, 1)$, $(3, -1)$, $(0, 0)$ for x_1, x_2. These are the rows of
$$S(\mathbf{x}) = \begin{bmatrix} 0 & 1 \\ 3 & -1 \\ 0 & 0 \end{bmatrix} \mathbf{x},$$
so S is a matrix transformation, hence linear.

P7.

Idea. Onto means a pivot in every row, one-to-one a pivot in every column (Thm. 12).

1. Forward phase:
$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 5 & 1 & 7 \\ 1 & 3 & 2 & 5 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$
2. Backward phase, $R_2 - R_3$ and then $R_1 - 2R_2$:
$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$
3. There is a pivot in every row, so T is onto $\mathbb{R}^3$. Column 4 has no pivot, so T is not one-to-one, as the size check predicts (4 columns, 3 rows).

P8.

Idea. Write $(3,-2) = 3\mathbf{e}_1 - 2\mathbf{e}_2$ and use superposition; the standard matrix has columns $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$.

1. By superposition,
$$T(3,-2) = T(3\mathbf{e}_1 - 2\mathbf{e}_2) = 3\begin{bmatrix}2\\-1\\3\end{bmatrix} - 2\begin{bmatrix}0\\4\\1\end{bmatrix} = \begin{bmatrix}6\\-11\\7\end{bmatrix}.$$

2. By Theorem 10, the standard matrix is
$$\begin{bmatrix}2 & 0\\-1 & 4\\3 & 1\end{bmatrix}.$$